

Anjali Kaushik

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EDUCATION

Arizona State University

Tempe, Arizona, USA

Master of Science, Computer Science, *August 2024 - Present*

GPA: 3.77/4

- Fall'24 Coursework - Deeplearning Applications, Image Processing & Analysis, Multimedia & Web Databases
- Spring'25 Coursework – Reinforcement Learning, Machine Learning Acceleration, Cloud Computing

Vellore Institute of Technology

Bhopal, Madhya Pradesh, India

Bachelor of Technology, Computer Science and Engineering, *July 2018 - May 2022*

GPA: 8.99/10

EXPERIENCE

Arizona State University

Tempe, AZ, USA

Knowledge Discovery & Data Mining Group - Research Assistant

March 2025 – Present

- Working on the applications of Reinforcement Learning in Causality. Training an agent to discover causal relationships in complex environments and datasets.
- Conducting a comparative study with different methods of measuring causal relationships and losses.

Shell PLC

Bengaluru, Karnataka, India

Associate Analytics Engineer

August 2022 – August 2024

- Worked on large complex financial data using Alteryx and SQL databases to automate workflows which resulted in 40% reduction in the workload of the team. Built an interactive dashboard to provide a holistic view of the finances.
- Collaborated with a global team on a deep learning-based image analysis project for lube monitoring. It involved developing a python-based model for ROI detection and classification. Worked with Linux clusters to enhance workflow efficiency.
- Automated geo-mechanical experiments with python which lead to a drastic reduction in the time to generate reports by 50% per lab technician.
- Developed a process automation framework for business-critical deployments for disaster recovery involving more than 100 applications. This led to a reduction in the workload of 3 business days per team member.
- Developed an interactive dashboard to track 150+ business continuity services, including disaster recovery and enterprise recovery for business-critical applications.
- Collaborated with 20+ application owners, stakeholders, and vendors to ensure the seamless execution of the disaster recovery process being a disaster recovery focal point.

PUBLICATIONS & PATENTS

Vellore Institute of Technology

Bhopal, Madhya Pradesh, India

Advanced Deepfake Detection using Inception-ResNet-v2

August 2021 – May 2022

- Published in ICCIS'23 by the Soft-Computing Society of India: Deepfake Detection using Inception-ResNet-V2 with an interactive GUI.- [Advanced Deepfake Detection Using Inception-ResNet-v2-paper-link](#)

Shell PLC

Bengaluru, Karnataka, India

MITRA (Patent Pending)

- Marine Inspection Tool for Rating and Assessment: A solution in lube monitoring used for image classification with 90% accuracy.

PROJECTS

Arizona State University

Tempe, Arizona, USA

Dynamic Programming in Cooperative Autonomous Assembly

January 2025 – Present

- Developing a novel offline reinforcement learning approach for multi-arm robotic assembly, overcoming geometric constraints like physics and potential collisions in complex structures like the Burr puzzle.
- Using multi-agent dynamic programming to optimize force application and coordination, enabling successful assembly of complex structures.

Implementation of Flash Attention in CUDA

January 2025 – Present

- Implementing Flash Attention for large language models (Llama 3), reducing inference latency and memory bandwidth through operator fusion and SoftMax tiling.
- Using Triton compiler to create attention modules, minimizing HBM transfers and enhancing memory access, resulting in improved throughput.

Cloud-Based Face Recognition System Architecture and Optimization

February 2025 – March 2025

- Designed a scalable cloud-based face recognition system on AWS using S3, SQS, and dynamic EC2 instances for efficient image processing.
- Built a responsive Python web tier to handle HTTP requests, manage cloud storage, and coordinate communication between the UI and processing layer.
- Engineered the application tier for deep learning inference with PyTorch, processing images from cloud storage and

returning results via asynchronous message queues.

- Developed an auto-scaling controller to dynamically provision resources, optimizing costs and ensuring system responsiveness with zero idle instances.

Self-Supervised Dense Point Tracking in Turbulent Videos

August 2024 – December 2024

- Developed a benchmark for point tracking in videos with induced atmospheric turbulence, enhancing the DINO-Tracker with RAFT-based optical flow refinement.
- Estimated the strength of turbulence using the trajectories of points in the turbulent videos.

Swin Transformer for Vision Tasks

August 2024 – December 2024

- Used Swin Transformer for classification, localization, and segmentation on ChestXRay14, NODE21, and ChexMask datasets to optimize pipelines and improve performance in medical image analysis.
- Achieved 72.64% accuracy training a model from scratch and 81.78% accuracy fine-tuning on ImageNet for ChestXRay14 classification.

Comparative Study of Video Retrieval Methods

August 2024 – December 2024

- Implemented dimensionality reduction techniques (PCA, SVD, LDA, K-Means) on ResNet, HOG, HOF, and Color-Histogram feature spaces to rank and retrieve videos based on similarity.
- Developed centroid-based ranking and label prediction models, achieving improved retrieval accuracy and demonstrating SVD's effectiveness in preserving latent structures.

TECHICAL SKILLS

Languages: Python, C++, MATLAB, R

Tools/Technologies: Linux, SQL, Alteryx, Flask, Git, Power Bi, GCP, AWS

Specializations: Computer Vision, Deep Learning, Image Processing, Reinforcement Learning, Cloud Computing, LLMs